Discriminative Project — Milestone 2: Object Detection (YOLOv8)

We combine existing single object photographs into 2×2 and 3×3 images, generating at least 100 images per student. Background only photographs are excluded.

We trained YOLOv8 to predict object IDs and bounding boxes, then evaluate it on unseen combinations from both layouts.

0. Setup

```
!pip install -q ultralytics

import os, io, json, random, shutil, time
from pathlib import Path
from collections import defaultdict

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as patches
from PIL import Image, ImageEnhance, ImageOps

SEED = 42
random.seed(SEED)
np.random.seed(SEED)

print("Setup complete.")
```

```
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Setup complete.
```

1. CONFIG

This section sets the dataset path, group size, number of generated images, image dimensions, and validation split. We generate at least 100 images per student, include both 2×2 and 3×3 layouts, and exclude background only photographs.

```
from google.colab import drive
drive.mount('/content/drive')
DATASET_ROOT = '/content/drive/MyDrive/IE7615 Deep Learning for AI'

GROUP_SIZE = 3
MIN_REQUIRED_IMAGES = GROUP_SIZE * 100
N_SYNTHETIC_IMAGES = max(400, MIN_REQUIRED_IMAGES)

OBJECT_COUNT_WEIGHTS = [
    (2, 1), (3, 1),
    (4, 5),   # 2x2 grid layout
    (5, 1), (6, 1), (7, 1), (8, 1),
    (9, 5),   # 3x3 grid layout
]
GRID_LAYOUT_THRESHOLD = 4

CANVAS_SIZE = 640
VAL_FRACTION = 0.15

OUT_DIR = Path('/content/drive/MyDrive/Discriminative_Project/milestor
OUT_DIR.mkdir(parents=True, exist_ok=True)

YOLO_DATA_DIR = Path('/content/yolo_dataset')

print(f"Minimum required images ({GROUP_SIZE} students x 100): {MIN_RI
print(f"Generating: {N_SYNTHETIC_IMAGES}")
```

```
Mounted at /content/drive
Minimum required images (3 students x 100): 300
Generating: 400
```

2. Copy dataset to local disk

```
LOCAL_DATASET_ROOT = '/content/dataset_local'

print(f"Copying dataset from Drive to local disk")
t0 = time.time()
!rm -rf "{LOCAL_DATASET_ROOT}"
!mkdir -p "{LOCAL_DATASET_ROOT}"
!cp -r "{DATASET_ROOT}/." "{LOCAL_DATASET_ROOT}/"
print(f"Done in {time.time()-t0:.1f}s")

DATASET_ROOT = LOCAL_DATASET_ROOT
```

```
Copying dataset from Drive to local disk
Done in 165.1s
```

3. Build the list of source images per class

This cell groups photographs by object ID, excludes background only and unreadable files, and removes exact duplicates. It then divides the photographs into training, validation, and test sets before generating composites, keeping source images separate across the three sets.

```
import hashlib

dataset_root = Path(DATASET_ROOT)
class_folders = sorted(
    p for p in dataset_root.iterdir()
    if p.is_dir() and p.name.startswith("images_")
)

class_images = {}
seen_hashes = {}
skipped_files = []

for folder in class_folders:
    obj_id = folder.name.replace("images_", "", 1)
    files = []

    for path in sorted(folder.glob("*")):
        if path.suffix.lower() not in (".jpg", ".jpeg", ".png"):
            continue
```

```
            if "_bg_" in path.stem.lower():
                continue

            try:
                with Image.open(path) as image:
                    image.verify()
                digest = hashlib.sha256(path.read_bytes()).hexdigest()
            except Exception as error:
                skipped_files.append((str(path), str(error)))
                continue

            if digest in seen_hashes:
                previous_id = seen_hashes[digest]
                if previous_id != obj_id:
                    raise ValueError(
                        f"Identical photograph assigned to different IDs:
                        f"{previous_id} and {obj_id}. Resolve before trai
                    )
                continue

            seen_hashes[digest] = obj_id
            files.append(path)

        if files:
            class_images[obj_id] = files

    class_names = sorted(class_images)
    class_to_index = {name: i for i, name in enumerate(class_names)}
    index_to_class = {i: name for name, i in class_to_index.items()}

    TEST_FRACTION = 0.15
    split_rng = random.Random(SEED)
    source_pools = {split: {} for split in ("train", "val", "test")}

    for obj_id, files in class_images.items():
        files = list(files)
        split_rng.shuffle(files)

        if len(files) < 7:
            raise ValueError(f"{obj_id} needs at least 7 unique source ph

        n_val_sources = max(1, int(len(files) * VAL_FRACTION))
        n_test_sources = max(1, int(len(files) * TEST_FRACTION))

        source_pools["val"][obj_id] = files[:n_val_sources]
```

```
        source_pools["test"][obj_id] = files[
            n_val_sources:n_val_sources + n_test_sources
        ]
        source_pools["train"][obj_id] = files[
            n_val_sources + n_test_sources:
        ]

    print(f"Classes: {len(class_names)}")
    print(f"Unique usable source photographs: {sum(map(len, class_images.v
    print(f"Unreadable files skipped: {len(skipped_files)}")
    print("Background only photographs excluded.")
    print("Source photographs are separated before generating composites."
```

```
Classes: 73
Unique usable source photographs: 6948
Unreadable files skipped: 0
Background only photographs excluded.
Source photographs are separated before generating composites.
```

4. Generate the synthetic multi object dataset

This section combines randomly selected single object photographs into 2×2 and 3×3 grids. Training images receive random horizontal flips and brightness and contrast adjustments. Each composite is saved with a matching YOLO label file containing class IDs and normalized bounding box coordinates, using photograph tile boundaries unless actual object annotations are provided.

Images and labels are written directly into the YOLO directory layout Ultralytics expects.

```
SOURCE_BOXES_FILE = OUT_DIR / "source_object_boxes.json"
source_object_boxes = (
    json.loads(SOURCE_BOXES_FILE.read_text())
    if SOURCE_BOXES_FILE.exists() else None
)

BOX_LABEL_MODE = (
    "actual_object_boxes"
    if source_object_boxes is not None
    else "photo_tile_boxes"
)
```

```
if source_object_boxes is not None:
    for obj_id, paths in class_images.items():
        for path in paths:
            key = path.relative_to(dataset_root).as_posix()
            if key not in source_object_boxes:
                raise ValueError(f"Missing source object annotation: ·

            with Image.open(path) as image:
                width, height = image.size

            x1, y1, x2, y2 = source_object_boxes[key]
            if not (0 <= x1 < x2 <= width and 0 <= y1 < y2 <= height)
                raise ValueError(f"Invalid source bounding box: {key}'
else:
    print(
        "WARNING: No source object annotations found. "
        "Labels and evaluation will describe photograph tiles, "
    )

def iou(box_a, box_b):
    ax1, ay1, ax2, ay2 = box_a
    bx1, by1, bx2, by2 = box_b
    intersection = (
        max(0, min(ax2, bx2) - max(ax1, bx1))
        * max(0, min(ay2, by2) - max(ay1, by1))
    )
    union = (
        (ax2 - ax1) * (ay2 - ay1)
        + (bx2 - bx1) * (by2 - by1)
        - intersection
    )
    return intersection / union if union > 0 else 0.0

def generate_one_image(
    canvas_size, weights=None, grid_threshold=None,
    split="train", grid_n=None, augment=None
):
    if grid_n is None:
        grid_n = random.choice((2, 3))
    if augment is None:
        augment = split == "train"

    edges = [
        round(i * canvas_size / grid_n)
```

```
            for i in range(grid_n + 1)
        ]
        canvas = Image.new("RGB", (canvas_size, canvas_size))
        labels, source_paths = [], []

        selected_ids = random.sample(class_names, grid_n * grid_n)

        for position, obj_id in enumerate(selected_ids):
            row, column = divmod(position, grid_n)
            left, top = edges[column], edges[row]
            right, bottom = edges[column + 1], edges[row + 1]
            tile_width, tile_height = right - left, bottom - top

            path = random.choice(source_pools[split][obj_id])
            with Image.open(path) as image:
                patch = image.convert("RGB")

            width, height = patch.size
            key = path.relative_to(dataset_root).as_posix()

            if source_object_boxes is None:
                x1, y1, x2, y2 = 0, 0, width, height
            else:
                x1, y1, x2, y2 = source_object_boxes[key]

            if augment:
                if random.random() < 0.5:
                    patch = ImageOps.mirror(patch)
                    x1, x2 = width - x2, width - x1

                patch = ImageEnhance.Brightness(patch).enhance(
                    random.uniform(0.85, 1.15)
                )
                patch = ImageEnhance.Contrast(patch).enhance(
                    random.uniform(0.85, 1.15)
                )

            patch = patch.resize(
                (tile_width, tile_height),
                Image.Resampling.BICUBIC
            )
            canvas.paste(patch, (left, top))

            labels.append((
                obj_id,
```

```
                left + x1 * tile_width / width,
                top + y1 * tile_height / height,
                left + x2 * tile_width / width,
                top + y2 * tile_height / height
            ))
            source_paths.append(key)

    return canvas, labels, source_paths

def boxes_to_yolo_lines(labels, class_to_index, canvas_size):
    lines = []
    for obj_id, x1, y1, x2, y2 in labels:
        assert 0 <= x1 < x2 <= canvas_size
        assert 0 <= y1 < y2 <= canvas_size

        lines.append(
            f"{class_to_index[obj_id]} "
            f"{(x1 + x2) / (2 * canvas_size):.8f} "
            f"{(y1 + y2) / (2 * canvas_size):.8f} "
            f"{(x2 - x1) / canvas_size:.8f} "
            f"{(y2 - y1) / canvas_size:.8f}"
        )
    return lines

print("2×2 and 3×3 generation functions ready.")
print(f"Bounding box label mode: {BOX_LABEL_MODE}")
```

```
WARNING: No source object annotations found. Labels and evaluation wil
2×2 and 3×3 generation functions ready.
Bounding box label mode: photo_tile_boxes
```

```
import yaml

YOLO_DATA_DIR = Path("/content/yolo_dataset_gapfree")
YOLO_DATA_DIR.mkdir(parents=True, exist_ok=True)

random.seed(SEED)
np.random.seed(SEED)

images_per_student = max(
    100, (N_SYNTHETIC_IMAGES + GROUP_SIZE - 1) // GROUP_SIZE
)
images_per_student += images_per_student % 2
N_SYNTHETIC_IMAGES = GROUP_SIZE * images_per_student
```

```
manifest = []
used_combinations = set()
t0 = time.time()

for split in ("train", "val", "test"):
    (YOLO_DATA_DIR / "images" / split).mkdir(parents=True, exist_ok=T
    (YOLO_DATA_DIR / "labels" / split).mkdir(parents=True, exist_ok=T

for student in range(1, GROUP_SIZE + 1):
    for grid_n in (2, 3):
        layout_count = images_per_student // 2
        n_val = max(1, round(layout_count * VAL_FRACTION))
        n_test = max(1, round(layout_count * TEST_FRACTION))
        counts = {
            "train": layout_count - n_val - n_test,
            "val": n_val,
            "test": n_test
        }

        for split, count in counts.items():
            for number in range(count):
                for attempt in range(1000):
                    canvas, labels, sources = generate_one_image(
                        CANVAS_SIZE, split=split, grid_n=grid_n
                    )
                    signature = tuple(sorted(sources))
                    if signature not in used_combinations:
                        used_combinations.add(signature)
                        break
                else:
                    raise RuntimeError("Could not generate a new sour

                stem = (
                    f"student{student}_{grid_n}x{grid_n}_"
                    f"{split}_{number:04d}"
                )
                canvas.save(
                    YOLO_DATA_DIR / "images" / split / f"{stem}.jpg",
                    quality=95
                )
                (YOLO_DATA_DIR / "labels" / split / f"{stem}.txt").wr
                    "\n".join(
                        boxes_to_yolo_lines(
                            labels, class_to_index, CANVAS_SIZE
```

```
                        )
                    ) + "\n"
                )
                manifest.append({
                    "image": f"{stem}.jpg",
                    "student": student,
                    "split": split,
                    "layout": f"{grid_n}x{grid_n}",
                    "sources": sources
                })

manifest_df = pd.DataFrame(manifest)

data_config = {
    "path": str(YOLO_DATA_DIR),
    "train": "images/train",
    "val": "images/val",
    "test": "images/test",
    "names": dict(enumerate(class_names))
}
(YOLO_DATA_DIR / "data.yaml").write_text(
    yaml.safe_dump(data_config, sort_keys=False)
)
shutil.copy(YOLO_DATA_DIR / "data.yaml", OUT_DIR / "data.yaml")

(OUT_DIR / "class_names.json").write_text(
    json.dumps(class_names, indent=2)
)
(OUT_DIR / "gapfree_manifest.json").write_text(
    json.dumps(manifest, indent=2)
)

print(f"Generated {len(manifest)} images in {time.time() - t0:.1f}s.")
print(f"Each student contributed {images_per_student} images.")
print(f"Annotation mode: {BOX_LABEL_MODE}")
display(pd.crosstab(
    [manifest_df["student"], manifest_df["split"]],
    manifest_df["layout"]
))
```

```
Generated 402 images in 11.9s.
Each student contributed 134 images.
Annotation mode: photo_tile_boxes
```

	layout	2x2	3x3
student	split		

student	split		
1	test	10	10
	train	47	47
	val	10	10
2	test	10	10
	train	47	47
	val	10	10
3	test	10	10
	train	47	47
	val	10	10

4b. Generate clean 2x2 and 3x3 samples for testing

This cell selects one held out test image from each layout, saves it with its matching label file, and displays the clean 2×2 and 3×3 samples without prediction boxes.

```
samples_dir = OUT_DIR / "ta_samples"
samples_dir.mkdir(parents=True, exist_ok=True)

for layout in ("2x2", "3x3"):
    row = manifest_df[
        (manifest_df["split"] == "test")
        & (manifest_df["layout"] == layout)
    ].iloc[0]

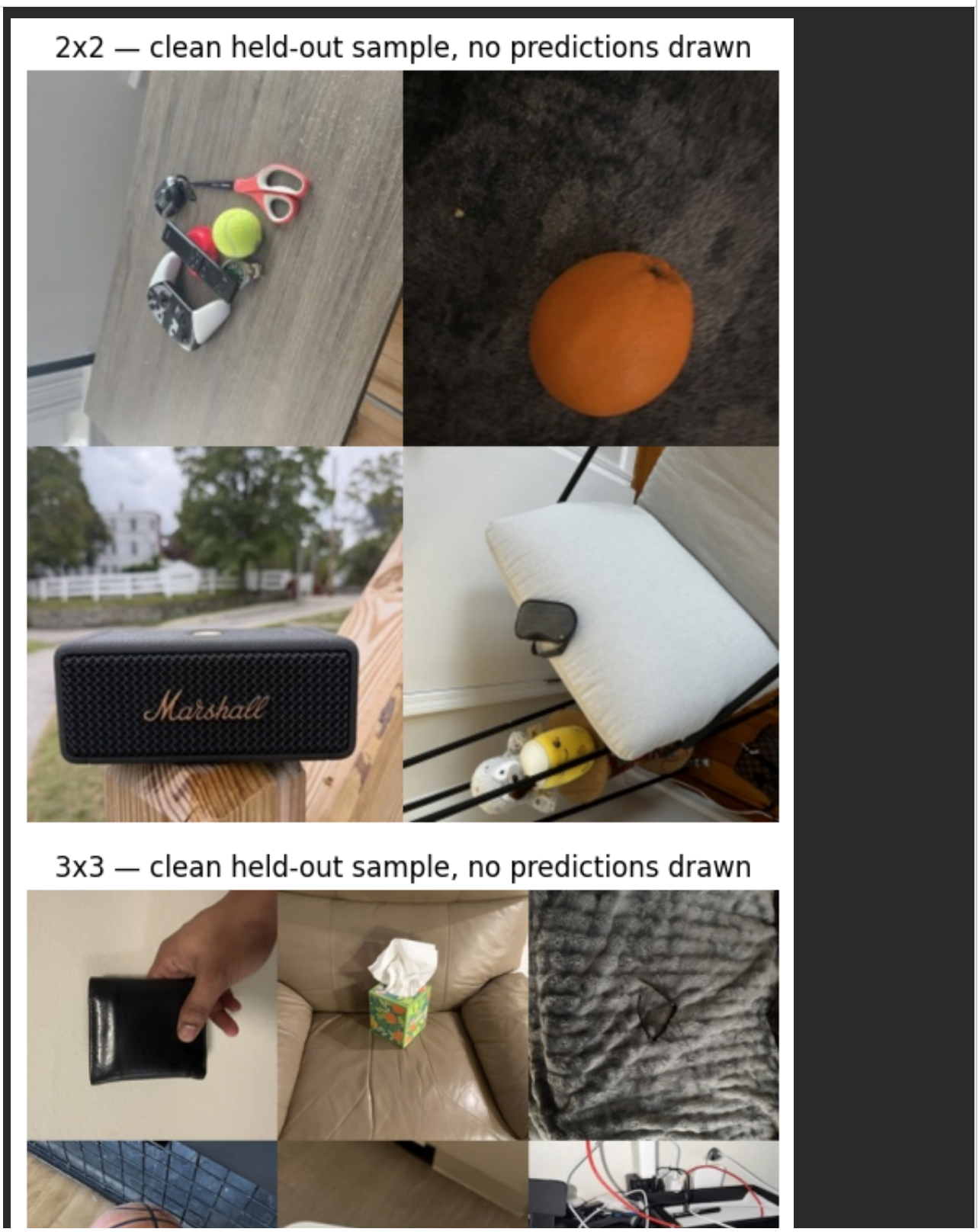
    image_path = YOLO_DATA_DIR / "images/test" / row["image"]
    label_path = (
        YOLO_DATA_DIR / "labels/test" / f"{image_path.stem}.txt"
    )

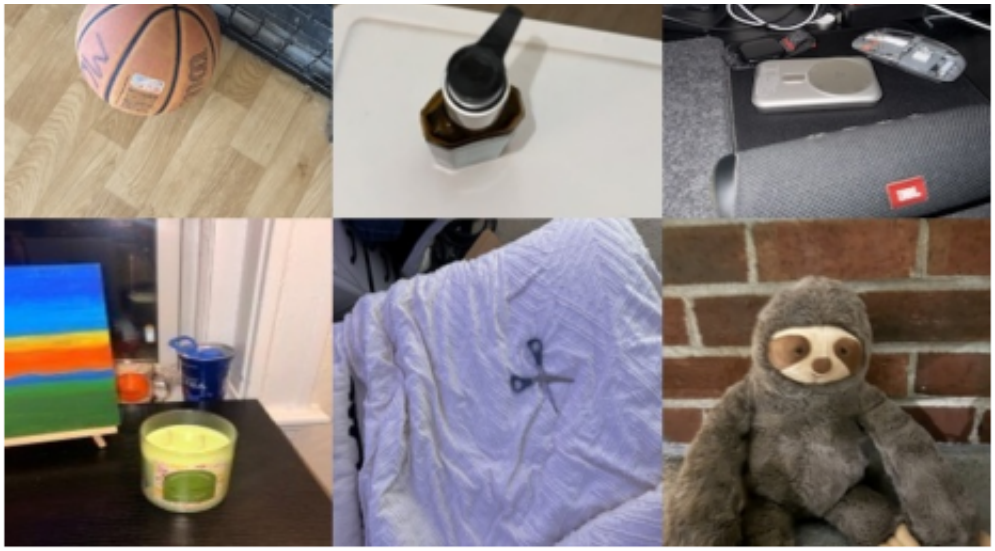
    shutil.copy(image_path, samples_dir / f"sample_{layout}.jpg")
    shutil.copy(label_path, samples_dir / f"sample_{layout}.txt")

    with Image.open(image_path) as image:
        plt.figure(figsize=(6, 6))
        plt.imshow(image)
```

```
            plt.title(f"{layout} — clean held-out sample, no predictions d
            plt.axis("off")
            plt.show()

print("Saved clean 2×2 and 3×3 samples.")
```





```
Saved clean 2×2 and 3×3 samples.
```

5. Sanity check a few generated images

Draw the recorded bounding boxes back onto a few synthetic training images, to confirm the labels actually line up with where objects were pasted before spending time training on them.

```
def read_ground_truth(img_path, label_path):
    with Image.open(img_path) as image:
        width, height = image.size

    objects = []
    for line in label_path.read_text().splitlines():
        cls_idx, xc, yc, w, h = map(float, line.split())
        cls_idx = int(cls_idx)
        assert 0 <= cls_idx < len(class_names)

        box = (
            (xc - w / 2) * width,
            (yc - h / 2) * height,
            (xc + w / 2) * width,
            (yc + h / 2) * height
        )
        objects.append((cls_idx, box))

    return objects
```

```
def show_sample_with_boxes(img_path, label_path, class_names, title=N
    with Image.open(img_path) as image:
        image = image.convert("RGB")

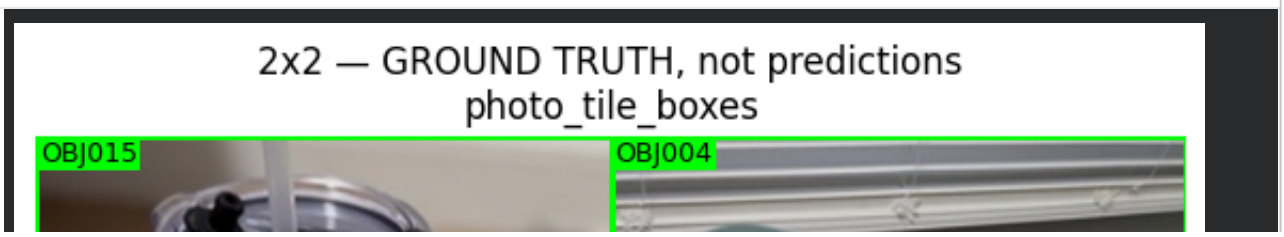
    fig, ax = plt.subplots(figsize=(7, 7))
    ax.imshow(image)

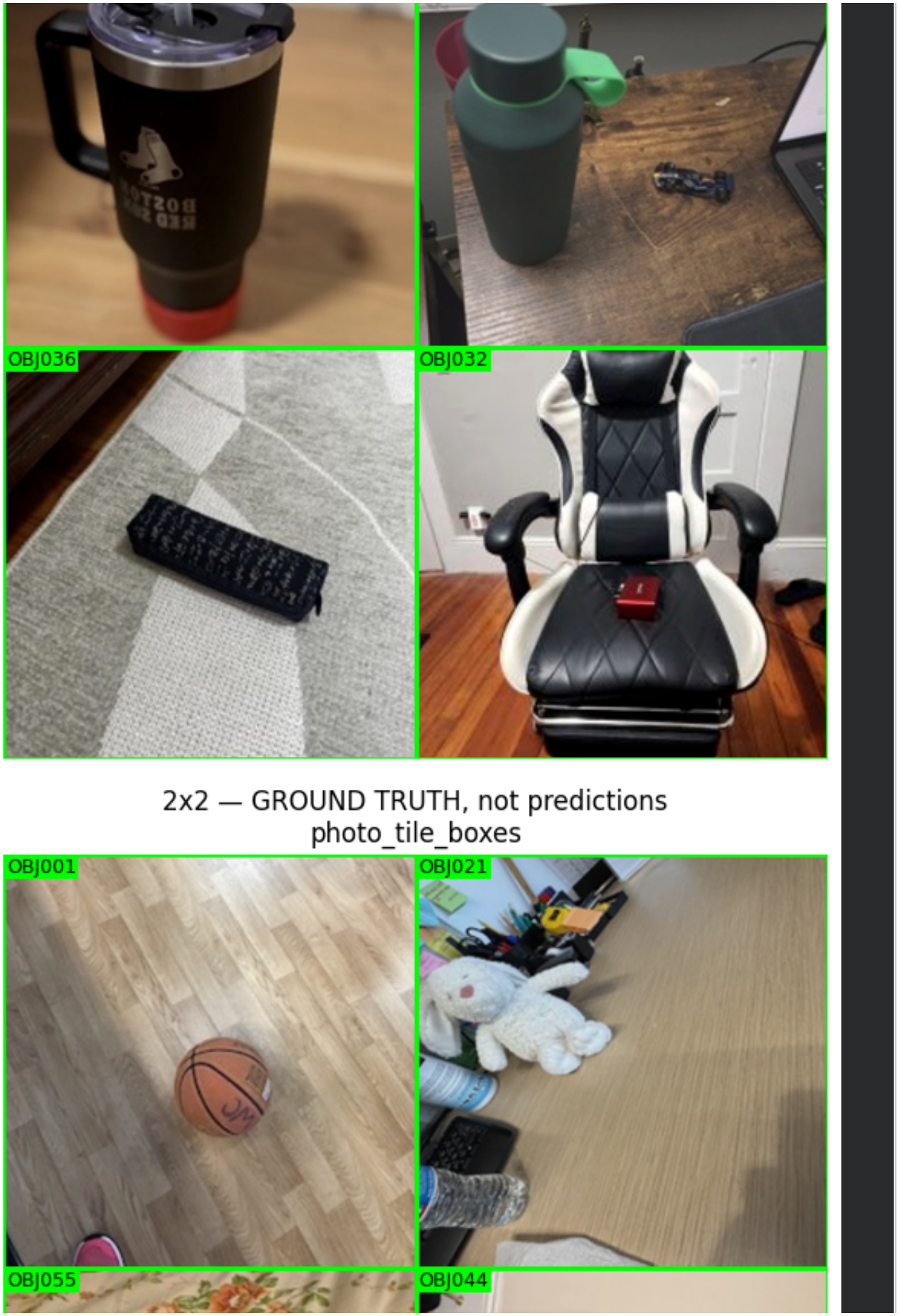
    for cls_idx, (x1, y1, x2, y2) in read_ground_truth(
        img_path, label_path
    ):
        ax.add_patch(patches.Rectangle(
            (x1, y1), x2 - x1, y2 - y1,
            linewidth=2, edgecolor="lime", facecolor="none"
        ))
        ax.text(
            x1 + 2, y1 + 2, class_names[cls_idx],
            va="top", color="black", fontsize=9,
            bbox={"facecolor": "lime", "edgecolor": "none", "pad": 1}
        )

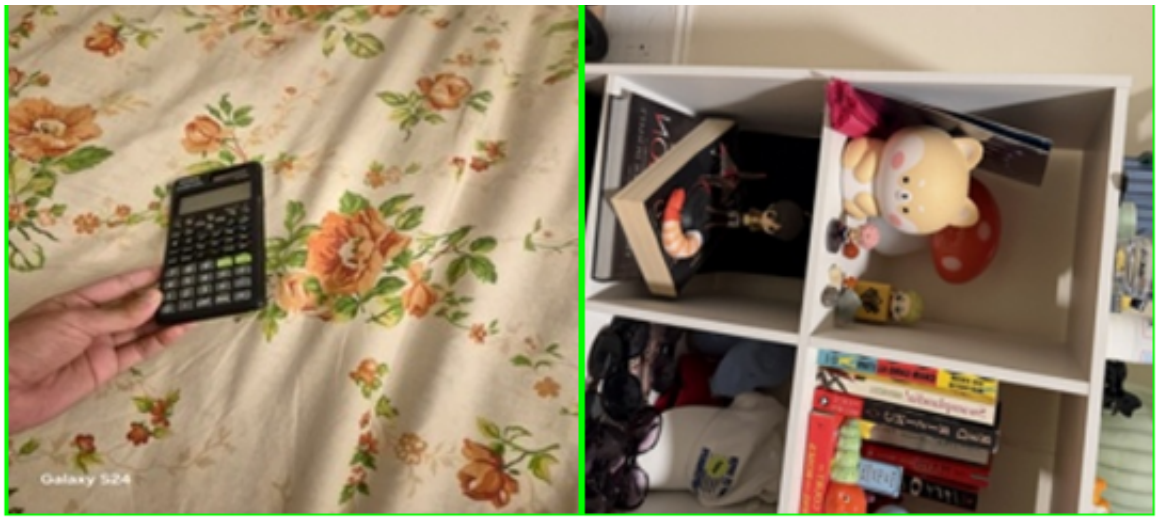
    ax.set_title(title or f"GROUND TRUTH — {BOX_LABEL_MODE}")
    ax.axis("off")
    plt.show()

for layout in ("2x2", "3x3"):
    rows = manifest_df[
        (manifest_df["split"] == "train")
        & (manifest_df["layout"] == layout)
    ].head(2)

    for filename in rows["image"]:
        image_path = YOLO_DATA_DIR / "images/train" / filename
        label_path = (
            YOLO_DATA_DIR / "labels/train" / f"{image_path.stem}.txt"
        )
        show_sample_with_boxes(
            image_path, label_path, class_names,
            title=f"{layout} — GROUND TRUTH, not predictions\n{BOX_LAI
        )
```




OBJ036
OBJ032
2x2 — GROUND TRUTH, not predictions
photo_tile_boxes
OBJ001
OBJ021
OBJ055
OBJ044




3x3 — GROUND TRUTH, not predictions
photo_tile_boxes
OBJ015
OBJ004
OBJ017
OBJ002
OBJ046
OBJ031
OBJ042
OBJ003
OBJ023


3x3 — GROUND TRUTH, not predictions
photo_tile_boxes
OBJ071
OBJ028
OBJ056
OBJ058
OBJ030
OBJ053
OBJ044
OBJ059
OBJ052

```
import logging
from ultralytics.utils import LOGGER

class RemoveLogEmojis(logging.Filter):
    def filter(self, record):
        message = record.getMessage()
        for symbol in ("🚀", "✅", "\ufe0f"):
            message = message.replace(symbol, "")
        record.msg = message
        record.args = ()
        return True

if not any(isinstance(f, RemoveLogEmojis) for f in LOGGER.filters):
    LOGGER.addFilter(RemoveLogEmojis())
```

```
Creating new Ultralytics Settings v0.0.8 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ul
Update Settings with 'yolo settings key=value', i.e. 'yolo settings ru
```

6. Train YOLOv8

This cell fine tunes the YOLOv8n model on the generated dataset for up to 50 epochs with a batch size of 16.After training, it loads the best checkpoint for evaluation and checks that the model's class names match the dataset.

```
from ultralytics import YOLO

EPOCHS = 50
IMG_SIZE = CANVAS_SIZE
BATCH_SIZE = 16

model = YOLO("yolov8n.pt")

results = model.train(
    data=str(YOLO_DATA_DIR / "data.yaml"),
    epochs=EPOCHS,
    imgsz=IMG_SIZE,
    batch=BATCH_SIZE,
    seed=SEED,
    project=str(OUT_DIR / "runs"),
    name="yolov8_gapfree",
```

```
    exist_ok=False,
    mosaic=0.0,
    mixup=0.0
)

best_weights = Path(model.trainer.save_dir) / "weights/best.pt"
detector = YOLO(str(best_weights))

assert [
    detector.names[i] for i in range(len(class_names))
] == class_names
```

OBJ014	6	6	1	0
OBJ015	4	4	0.656	0.75
OBJ016	6	6	0.498	0.333
OBJ017	11	11	1	0.673
OBJ018	7	7	1	0.963
OBJ019	6	6	0.912	1
OBJ020	4	4	1	0.462
OBJ021	7	7	0.899	0.714
OBJ022	5	5	0.609	0.8
OBJ023	11	11	1	0.851
OBJ024	6	6	0.905	1
OBJ025	9	9	0.978	0.778
OBJ026	7	7	0.728	1
OBJ027	6	6	0.657	1
OBJ028	5	5	0.641	0.8
OBJ029	4	4	0.779	1
OBJ030	6	6	0.865	1
OBJ031	7	7	1	0.411
OBJ032	3	3	0.708	1
OBJ033	8	8	1	0.979
OBJ034	6	6	1	0
OBJ035	2	2	0.539	1
OBJ036	7	7	0.815	0.714
OBJ037	10	10	1	0.876
OBJ038	2	2	0.318	0.5
OBJ039	7	7	0.92	1
OBJ040	5	5	0.9	0.8
OBJ041	4	4	0.802	0.75
OBJ042	6	6	0.974	0.5
OBJ043	2	2	0.467	1
OBJ044	2	2	0.745	1
OBJ045	3	3	0.228	0.228
OBJ046	3	3	0.908	1
OBJ047	3	3	0.817	0.667
OBJ048	6	6	0.861	0.667
OBJ049	6	6	0.938	1
OBJ050	9	9	1	0.378

OBJ051	12	12	0.912	1
OBJ052	3	3	0.721	0.884
OBJ053	4	4	1	0.973
OBJ054	6	6	0.748	0.667
OBJ055	2	2	0.766	1
OBJ056	6	6	1	0.714
OBJ057	8	8	1	0.22
OBJ058	5	5	0.777	1
OBJ059	7	7	0.949	0.429
OBJ060	3	3	0.739	1
OBJ061	6	6	0.748	0.667
OBJ062	2	2	0.182	0.365
OBJ063	3	3	1	0.599
OBJ064	5	5	0.927	0.8
OBJ065	8	8	0.989	0.75
OBJ066	4	4	0.764	1
OBJ067	5	5	0.759	1
OBJ068	7	7	0.636	0.143
OBJ069	3	3	1	0.64
OBJ070	2	2	0.797	1
OBJ071	2	2	0.4	0.5
OBJ072	5	5	0.922	1

7. Evaluate on the validation set

This cell evaluates the model separately on held out 2×2 and 3×3 images, reporting mAP, precision, recall, matched object rate, and exact image accuracy. Predictions count as correct when the class ID matches and IoU is at least 0.50, using a confidence threshold of 0.25.

```
CONF_THRESHOLD = 0.25
MATCH_IOU_THRESHOLD = 0.50

def score_prediction(prediction, ground_truth):
    predicted = [
        (int(cls), float(conf), box.tolist())
        for cls, conf, box in zip(
            prediction.boxes.cls.cpu().numpy(),
            prediction.boxes.conf.cpu().numpy(),
            prediction.boxes.xyxy.cpu().numpy()
        )
    ]
    predicted.sort(key=lambda item: item[1], reverse=True)
```

```
        matched = set()
        tp = 0

        for pred_cls, confidence, pred_box in predicted:
            candidates = [
                (iou(pred_box, gt_box), index)
                for index, (gt_cls, gt_box) in enumerate(ground_truth)
                if index not in matched and pred_cls == gt_cls
            ]
            if candidates:
                overlap, index = max(candidates)
                if overlap >= MATCH_IOU_THRESHOLD:
                    matched.add(index)
                    tp += 1

        fp = len(predicted) - tp
        fn = len(ground_truth) - tp
        exact = int(fp == 0 and fn == 0)
        return tp, fp, fn, exact

    evaluation_rows = []
    per_image_rows = []
    prediction_cache = {}

    for layout in ("2x2", "3x3"):
        filenames = manifest_df[
            (manifest_df["split"] == "test")
            & (manifest_df["layout"] == layout)
        ]["image"].tolist()

        image_paths = [
            YOLO_DATA_DIR / "images/test" / filename
            for filename in filenames
        ]

        image_list = YOLO_DATA_DIR / f"test_{layout}.txt"
        image_list.write_text(
            "\n".join(str(path.resolve()) for path in image_paths) + "\n"
        )

        layout_config = dict(data_config)
        layout_config["test"] = str(image_list)
        layout_yaml = YOLO_DATA_DIR / f"data_{layout}.yaml"
        layout_yaml.write_text(
            yaml.safe_dump(layout_config, sort_keys=False)
```

```
        )

        metrics = detector.val(
            data=str(layout_yaml),
            split="test",
            imgsz=IMG_SIZE,
            conf=0.001,
            project=str(OUT_DIR / "layout_evaluation"),
            name=f"test_{layout}",
            plots=True
        )

        totals = np.zeros(4, dtype=int)

        for image_path in image_paths:
            prediction = detector.predict(
                source=str(image_path),
                imgsz=IMG_SIZE,
                conf=CONF_THRESHOLD,
                verbose=False
            )[0]
            ground_truth = read_ground_truth(
                image_path,
                YOLO_DATA_DIR / "labels/test" / f"{image_path.stem}.txt"
            )

            tp, fp, fn, exact = score_prediction(prediction, ground_truth
            totals += [tp, fp, fn, exact]
            prediction_cache[image_path.name] = prediction

            per_image_rows.append({
                "image": image_path.name,
                "layout": layout,
                "TP": tp,
                "FP": fp,
                "FN": fn,
                "matched_object_rate_pct": 100 * tp / len(ground_truth),
                "exact_image_correct": exact
            })

        tp, fp, fn, exact = map(int, totals)
        evaluation_rows.append({
            "layout": layout,
            "annotation_mode": BOX_LABEL_MODE,
            "test_images": len(image_paths),
```

```
            "mAP50": float(metrics.box.map50),
            "mAP50_95": float(metrics.box.map),
            "precision_maxF1": float(metrics.box.mp),
            "recall_maxF1": float(metrics.box.mr),
            "precision_conf025": tp / max(1, tp + fp),
            "matched_object_rate_pct": 100 * tp / max(1, tp + fn),
            "exact_image_accuracy_pct": 100 * exact / len(image_paths),
            "TP": tp,
            "FP": fp,
            "FN": fn
        })

    evaluation_df = pd.DataFrame(evaluation_rows)
    per_image_df = pd.DataFrame(per_image_rows)

    evaluation_df.to_csv(OUT_DIR / "layout_test_metrics.csv", index=False
    per_image_df.to_csv(OUT_DIR / "per_image_test_metrics.csv", index=Fal

    eval_summary = {
        "annotation_mode": BOX_LABEL_MODE,
        "confidence_threshold": CONF_THRESHOLD,
        "matching_IoU_threshold": MATCH_IOU_THRESHOLD,
        "layouts": evaluation_rows
    }
    (OUT_DIR / "eval_summary.json").write_text(
        json.dumps(eval_summary, indent=2)
    )

    display(evaluation_df.round(4))
    print("Test composites use held out source photographs and different
    if BOX_LABEL_MODE == "photo_tile_boxes":
        print("photograph tile detection results.")
```

```
Ultralytics 8.4.175  Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla
Model summary (fused): 73 layers, 3,150,203 parameters, 0 gradients,
val: Fast image access  (ping: 0.0±0.0 ms, read: 2014.1±526.3 MB/s, s
val: Scanning /content/yolo_dataset_gapfree/labels/test... 30 images,
val: New cache created: /content/yolo_dataset_gapfree/labels/test.cac
                Class     Images  Instances      Box(P          R
                  all         30        120      0.696      0.735
               OBJ001          2          2      0.957          1
               OBJ002          4          4      0.822          1
               OBJ003          3          3      0.716      0.667
               OBJ005          3          3        0.8          1
               OBJ006          2          2      0.655          1
               OBJ007          3          3          1      0.949
               OBJ009          4          4      0.759          1
```

OBJ009	4	4	0.759	1
OBJ010	1	1	0.433	1
OBJ011	1	1	0.633	1
OBJ012	1	1	1	0
OBJ013	3	3	0.916	0.333
OBJ014	1	1	1	0
OBJ015	2	2	0.759	1
OBJ016	1	1	1	0
OBJ017	1	1	1	0
OBJ018	4	4	0.866	1
OBJ019	1	1	0.647	1
OBJ020	1	1	0.556	1
OBJ021	1	1	0.721	1
OBJ022	1	1	0.37	1
OBJ023	1	1	1	0
OBJ024	1	1	0.639	1
OBJ025	3	3	0.775	1
OBJ026	1	1	0.551	1
OBJ027	2	2	0.889	1
OBJ028	3	3	0.672	1
OBJ029	1	1	0.644	1
OBJ030	1	1	0.626	1
OBJ031	2	2	0.426	0.5
OBJ032	3	3	0.799	1
OBJ033	1	1	0.644	1
OBJ035	4	4	0.749	1
OBJ036	4	4	0.317	0.25
OBJ037	2	2	0.632	0.5
OBJ038	1	1	0	0
OBJ039	1	1	0.623	1
OBJ040	2	2	0.572	1
OBJ042	1	1	0.679	1
OBJ043	1	1	1	0
OBJ044	2	2	0.755	1
OBJ045	1	1	0.536	1
OBJ047	1	1	0.472	1
OBJ048	3	3	0.93	1
OBJ049	3	3	0.681	1
OBJ050	3	3	0.417	0.333
OBJ052	1	1	0.656	1
OBJ053	1	1	0.628	1
OBJ054	2	2	0.475	1
OBJ056	1	1	0.61	1
OBJ057	3	3	1	0.498
OBJ058	1	1	0.648	1
OBJ059	2	2	0.0356	0.0356
OBJ060	4	4	0.658	0.5
OBJ061	2	2	0.87	0.5
OBJ062	1	1	0	0
OBJ063	2	2	0.561	1
OBJ064	1	1	1	0

```
                OBJ065          1          1          1          0
                OBJ067          2          2      0.857          1
                OBJ069          4          4      0.892          1
                OBJ070          2          2      0.954        0.5
                OBJ071          2          2      0.692          1
Speed: 6.2ms preprocess, 18.0ms inference, 0.0ms loss, 6.2ms postproc
Results saved to /content/drive/MyDrive/Discriminative_Project/milest
Ultralytics 8.4.175  Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla
Model summary (fused): 73 layers, 3,150,203 parameters, 0 gradients,
val: Fast image access  (ping: 0.0±0.0 ms, read: 3251.0±679.7 MB/s, s
val: Scanning /content/yolo_dataset_gapfree/labels/test... 30 images,
val: New cache created: /content/yolo_dataset_gapfree/labels/test.cac
                 Class     Images  Instances      Box(P          R
                   all         30        270      0.856      0.688
                OBJ001          3          3      0.896          1
                OBJ002          1          1      0.754          1
                OBJ003          5          5          1      0.526
                OBJ004          3          3          1       0.49
                OBJ005          5          5      0.915          1
                OBJ006          6          6      0.988      0.833
                OBJ007          5          5      0.814        0.4
                OBJ008          6          6          1      0.952
                OBJ010          6          6          1      0.447
                OBJ011          6          6      0.892      0.667
                OBJ012          5          5      0.737        0.2
                OBJ013          1          1          0          0
                OBJ014          3          3      0.554      0.333
                OBJ015          3          3      0.715      0.333
                OBJ016          5          5          1      0.549
                OBJ017          7          7      0.951          1
                OBJ018          4          4      0.932          1
                OBJ019          3          3      0.917          1
                OBJ020          4          4      0.928          1
                OBJ021          2          2      0.877          1
                OBJ022          2          2      0.869          1
                OBJ023          1          1        0.7          1
                OBJ024          6          6      0.946          1
                OBJ025          4          4      0.875          1
                OBJ026          3          3      0.815      0.667
                OBJ027          3          3          1      0.484
                OBJ028          3          3      0.886          1
                OBJ029          5          5      0.938          1
                OBJ030          4          4      0.877       0.75
                OBJ031          2          2      0.927        0.5
                OBJ032          5          5          1      0.578
                OBJ033          1          1      0.749          1
                OBJ034          2          2          1          0
                OBJ035          3          3      0.744          1
                OBJ036          1          1          1          0
                OBJ037          3          3      0.879          1
```

```
                OBJ038           4           4       0.851        0.25
                OBJ039           8           8           1       0.959
                OBJ040           4           4       0.731        0.75
                OBJ041           3           3       0.871           1
                OBJ042           3           3           1       0.761
                OBJ043           3           3           1        0.87
                OBJ044           5           5           1       0.916
                OBJ045           4           4           1           0
                OBJ046           4           4           1       0.773
                OBJ047           2           2       0.819           1
                OBJ048           5           5       0.792           1
                OBJ049           3           3       0.895           1
                OBJ050           4           4           1           0
                OBJ051           3           3           1       0.731
                OBJ052           3           3       0.939       0.333
                OBJ053           3           3       0.883           1
                OBJ054           3           3       0.983       0.667
                OBJ055           4           4       0.713        0.75
                OBJ056           7           7           1       0.731
                OBJ057           2           2      0.0929      0.0929
                OBJ058           2           2       0.451           1
                OBJ059           4           4           1           0
                OBJ060           3           3        0.83       0.667
                OBJ061           2           2       0.789         0.5
                OBJ062           5           5        0.87         0.4
                OBJ063           4           4           1       0.728
                OBJ064           4           4       0.202       0.151
                OBJ065           5           5       0.889         0.4
                OBJ066           4           4       0.909         0.5
                OBJ067           7           7       0.841       0.759
                OBJ068           6           6       0.795       0.167
                OBJ069           2           2       0.881           1
                OBJ070           1           1       0.775           1
                OBJ071           4           4       0.907           1
                OBJ072           4           4       0.919           1
                OBJ073           5           5       0.912           1
Speed: 9.8ms preprocess, 5.0ms inference, 0.0ms loss, 2.3ms postproce
Results saved to /content/drive/MyDrive/Discriminative_Project/milest
```

	layout	annotation_mode	test_images	mAP50	mAP50_95	precision_ma
0	2x2	photo_tile_boxes	30	0.8060	0.8050	0.6
1	3x3	photo_tile_boxes	30	0.7797	0.7775	0.8

```
Test composites use held out source photographs and different combina
photograph tile detection results.
```

8. Run inference on sample images and visualize

Selects a few test images from both 2×2 and 3×3 layouts and displays the model's predicted object IDs and bounding boxes. These images serve as the sample detection results for the report.

```
inference_out_dir = OUT_DIR / "sample_detections_gapfree"
inference_out_dir.mkdir(parents=True, exist_ok=True)

predicted_locations = []

for layout in ("2x2", "3x3"):
    candidates = per_image_df[per_image_df["layout"] == layout]
    selected = candidates.sample(
        n=min(6, len(candidates)),
        random_state=SEED
    )

    for _, row in selected.iterrows():
        image_path = YOLO_DATA_DIR / "images/test" / row["image"]
        prediction = prediction_cache[image_path.name]

        annotated_rgb = prediction.plot(
            labels=True, conf=True, line_width=2
        )[:, :, ::-1]

        Image.fromarray(annotated_rgb).save(
            inference_out_dir / f"predicted_{image_path.name}"
        )

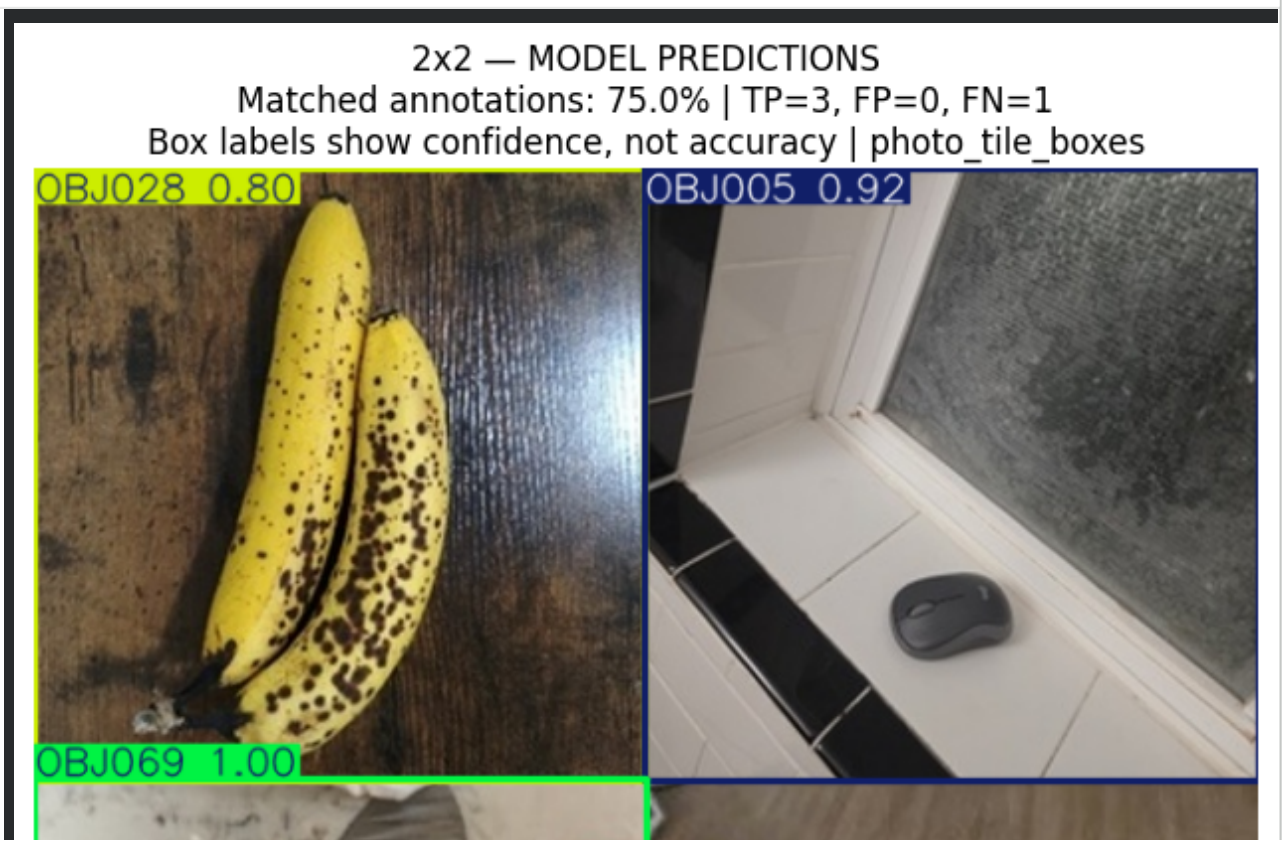
        for cls, confidence, box in zip(
            prediction.boxes.cls.cpu().numpy(),
            prediction.boxes.conf.cpu().numpy(),
            prediction.boxes.xyxy.cpu().numpy()
        ):
            predicted_locations.append({
                "image": image_path.name,
                "layout": layout,
                "object_id": detector.names[int(cls)],
                "confidence": float(confidence),
                "x_min": float(box[0]),
                "y_min": float(box[1]),
```

```
                    "x_max": float(box[2]),
                    "y_max": float(box[3])
                })

        plt.figure(figsize=(8, 8))
        plt.imshow(annotated_rgb)
        plt.title(
            f"{layout} — MODEL PREDICTIONS\n"
            f"Matched annotations: {row['matched_object_rate_pct']:.1
            f"TP={row['TP']}, FP={row['FP']}, FN={row['FN']}\n"
            f"Box labels show confidence, not accuracy | {BOX_LABEL_M
        )
        plt.axis("off")
        plt.show()

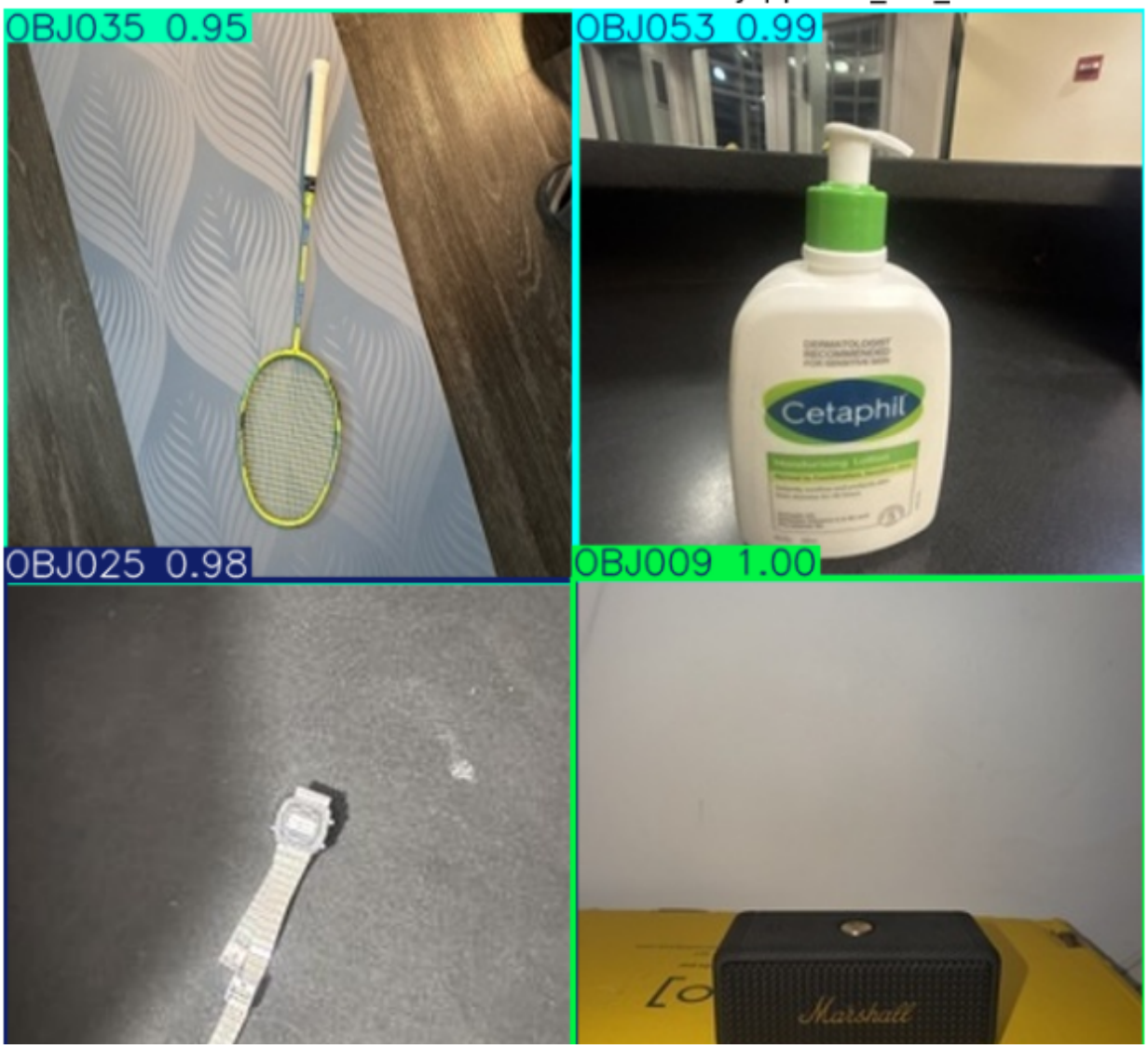
pd.DataFrame(
    predicted_locations,
    columns=[
        "image", "layout", "object_id", "confidence",
        "x_min", "y_min", "x_max", "y_max"
    ]
).to_csv(OUT_DIR / "predicted_ids_locations.csv", index=False)

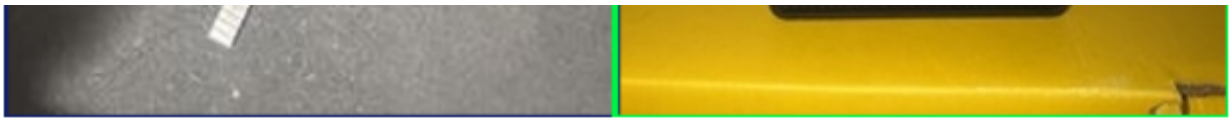
print("Saved predictions for different held out combinations of both
```



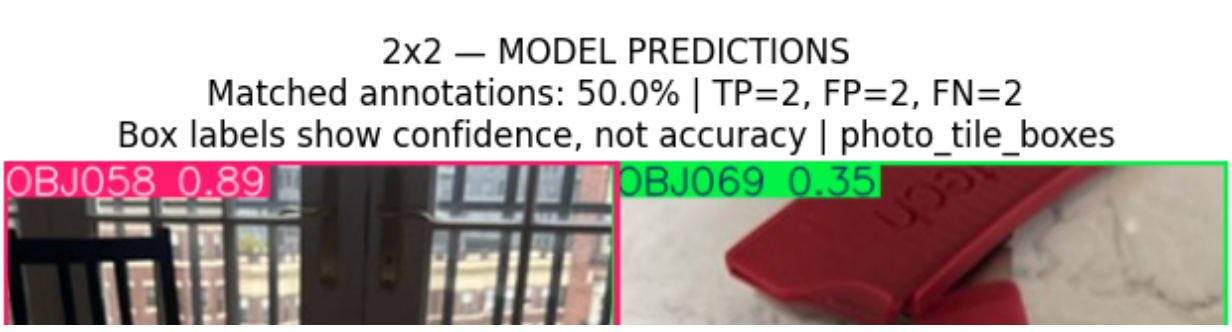


2x2 — MODEL PREDICTIONS
Matched annotations: 100.0% | TP=4, FP=0, FN=0
Box labels show confidence, not accuracy | photo_tile_boxes





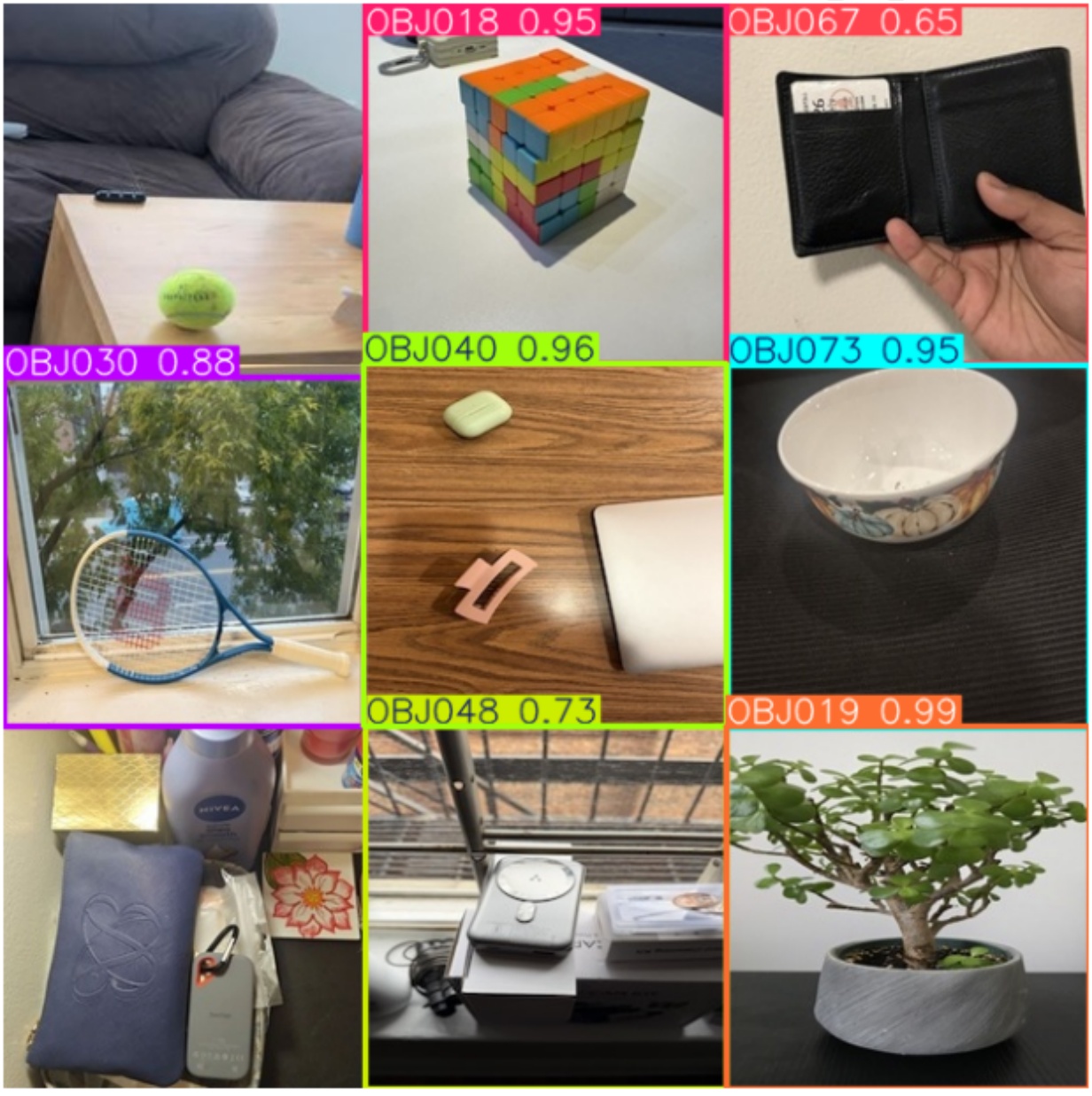

2x2 — MODEL PREDICTIONS
Matched annotations: 75.0% | TP=3, FP=0, FN=1
Box labels show confidence, not accuracy | photo_tile_boxes
OBJ063 0.88
OBJ009 0.35
OBJ011 0.99
Marshall

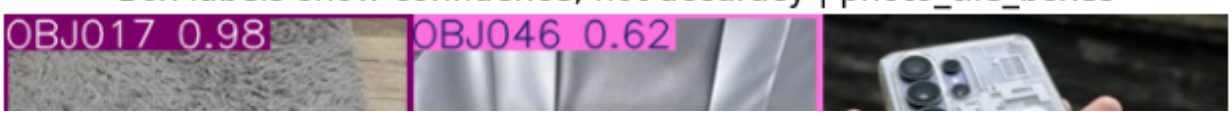

2x2 — MODEL PREDICTIONS
Matched annotations: 50.0% | TP=2, FP=2, FN=2
Box labels show confidence, not accuracy | photo_tile_boxes
OBJ058 0.89
OBJ069 0.35


OBJ040 0.45
OBJ024 1.00
Agilent
2x2 — MODEL PREDICTIONS
Matched annotations: 100.0% | TP=4, FP=0, FN=0
Box labels show confidence, not accuracy | photo_tile_boxes
OBJ009 1.00
OBJ071 0.99
OBJ027 0.98
OBJ028 0.95


2x2 — MODEL PREDICTIONS
Matched annotations: 75.0% | TP=3, FP=0, FN=1
Box labels show confidence, not accuracy | photo_tile_boxes
OBJ019 0.99
OBJ001 0.99
OBJ036 0.65




3x3 — MODEL PREDICTIONS
Matched annotations: 77.8% | TP=7, FP=0, FN=2
Box labels show confidence, not accuracy | photo_tile_boxes
OBJ018 0.95
OBJ067 0.65
OBJ030 0.88
OBJ040 0.96
OBJ073 0.95
OBJ048 0.73
OBJ019 0.99


3x3 — MODEL PREDICTIONS
Matched annotations: 66.7% | TP=6, FP=3, FN=3
Box labels show confidence, not accuracy | photo_tile_boxes
OBJ017 0.98
OBJ046 0.62

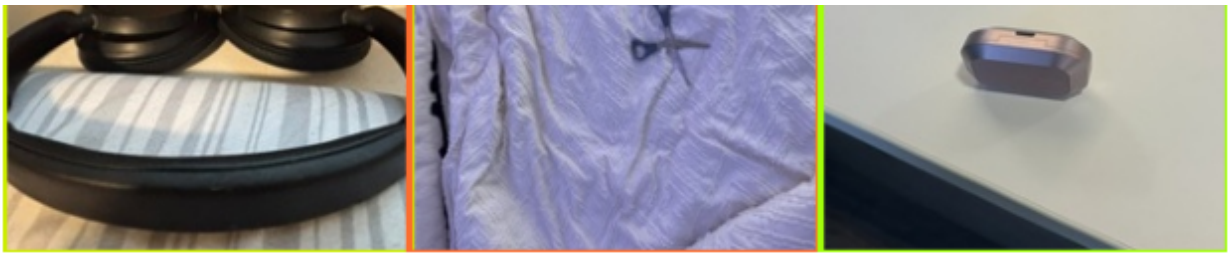


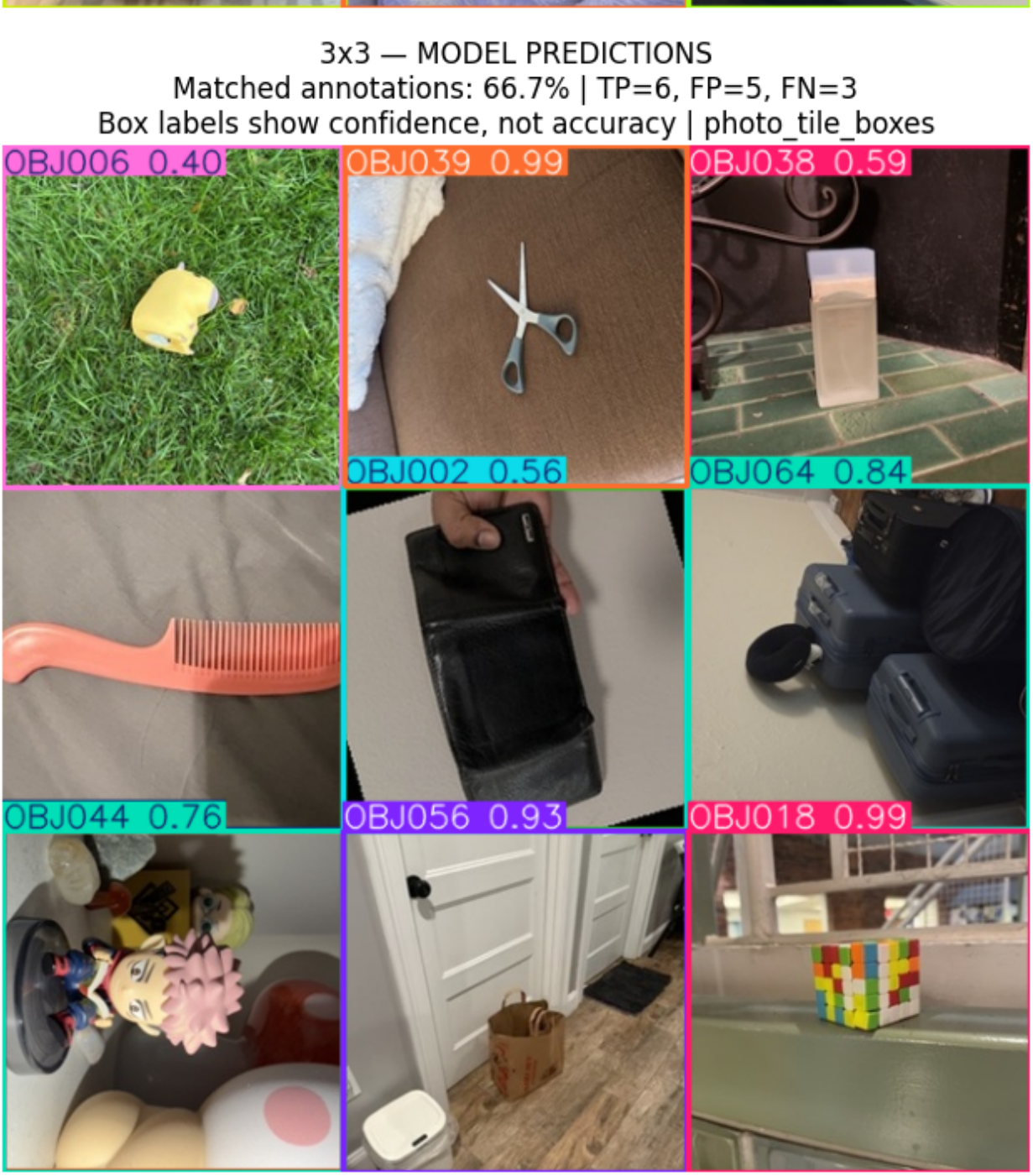
3x3 — MODEL PREDICTIONS
Matched annotations: 44.4% | TP=4, FP=1, FN=5
Box labels show confidence, not accuracy | photo_tile_boxes

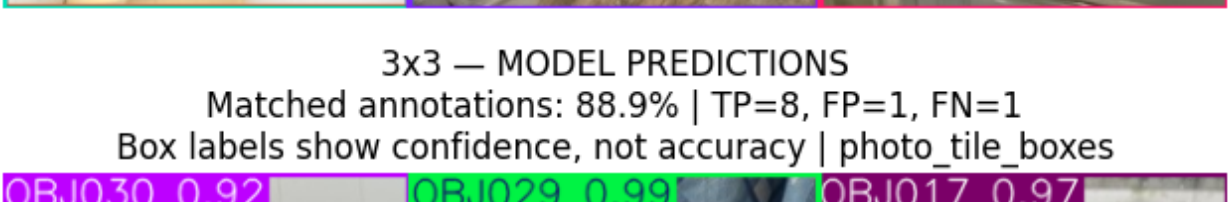



OBJ018 0.95


3x3 — MODEL PREDICTIONS
Matched annotations: 66.7% | TP=6, FP=3, FN=3
Box labels show confidence, not accuracy | photo_tile_boxes
OBJ064 0.93
OBJ011 0.99
OBJ060 1.00
OBJ056 0.55
OBJ072 0.99
OBJ048 0.57
OBJ039 0.81
OBJ020 0.99




3x3 — MODEL PREDICTIONS
Matched annotations: 66.7% | TP=6, FP=5, FN=3
Box labels show confidence, not accuracy | photo_tile_boxes
OBJ006 0.40
OBJ039 0.99
OBJ038 0.59
OBJ002 0.56
OBJ064 0.84
OBJ044 0.76
OBJ056 0.93
OBJ018 0.99


3x3 — MODEL PREDICTIONS
Matched annotations: 88.9% | TP=8, FP=1, FN=1
Box labels show confidence, not accuracy | photo_tile_boxes




```
Saved predictions for different held out combinations of both layouts.
```



9. Save the trained model

```
best_weights = Path(model.trainer.save_dir) / "weights" / "best.pt"

if best_weights.exists():
    shutil.copy(best_weights, OUT_DIR / "yolov8_best.pt")
    print("Best model saved.")
else:
    print("Best model weights not found.")
```

```
Best model saved.
```